7. A cow is tethered with a rope of length 35 m at the centre of the rectangular field of length 76 m and breadth 60 m. Find the area of the land that the cow cannot graze?

8. A path 5 m wide runs along the inside of the rectangular field. The length of the rectangular field is three times the breadth of the field. If the area of the path is 500 m^2 then find the length and breadth of the field.

9. A circular path has to be constructed around a circular ground. If the areas of the outer and inner circles are 1386 m^2 and 616 m^2 respectively, find the width and area of the path.

10. A goat is tethered with a rope of length 45 m at the centre of the circular grass land whose radius is 52 m. Find the area of the grass land that the goat cannot graze.

11. A strip of 4 cm wide is cut and removed from all the sides of the rectangular cardboard with dimensions 30 $cm \times 20$ cm. Find the area of the removed portion and area of the remaining cardboard.

12. A rectangular field is of dimension 20 $m \times 15$ m. Two paths run parallel to the sides of the rectangle through the centre of the field. The width of the longer path is 2 m and that of the shorter path is 1 m. Find (i) the area of the paths (ii) the area of the remaining portion of the field (iii) the cost of constructing the roads at the rate of ₹10 per $sq.m$.

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Summary

- Circle is a round plane figure whose boundary (the circumference) consists of points equidistant from the fixed point (the centre).
- Distance around the circular region is called the circumference or perimeter of the circle.
- Circumference of a circle, $C = \pi d$ $units$, where 'd' is the diameter of a circle and $\pi = \frac{22}{7}$ or 3.14 approximately.
- Area of the circle is the region enclosed by the circle.
- The area of the circle, $(A) = \pi r^2$ $sq.\ units$, where 'r' is the radius of the circle.
- Area of the circular path = Area of the outer circle – Area of the inner circle
  $= \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$ sq. units, where 'R' and 'r' are radius of the outer and inner circles respectively.
- Area of the rectangular path = Area of the outer rectangle – Area of the inner rectangle
  $= (LB - lb)$ $sq.units$, where L, B, and l, b are length and breadth of outer and inner rectangles respectively.

7th_Maths_T2_EM_Chp2.indd 40 09-07-2019 12:20:01